A Hybrid Approach for Image Enhancement and Visibility Restoration Under Atmospheric Degradation

Saikiran Hari
Department of Electronics and Communication Engineering
Potti Sriramulu Chalavadi Mallikarjuna Rao College Of Engineering And Technology
Vijayawada, Andhra Pradesh
harisaikiran17@gmail.com

Kundan Vadegharu
Department of Electronics and Communication Engineering
Potti Sriramulu Chalavadi Mallikarjuna Rao College Of Engineering And Technology
Vijayawada, Andhra Pradesh
kundanvadeghar@gmail.com

Chanikya Nandini Kolluri
Department of Electronics and Communication Engineering
Potti Sriramulu Chalavadi Mallikarjuna Rao College Of Engineering And Technology
Vijayawada, Andhra Pradesh
kollurinandini24@gmail.com

Shanmukha Sai Yadav Mugada
Department of Electronics and Communication Engineering
Potti Sriramulu Chalavadi Mallikarjuna Rao College Of Engineering And Technology
Vijayawada, Andhra Pradesh
mugadashanmukhasai@gmail.com

Pooja A Nair*
Department of Electronics and Communication Engineering
Potti Sriramulu Chalavadi Mallikarjuna Rao College Of Engineering And Technology
Vijayawada, Andhra Pradesh
poojaanair@pscmr.ac.in

***Abstract*—Image visibility degradation caused by atmospheric conditions such as fog, smoke, and sand Dust significantly affects the quality of captured images by reducing contrast, distorting colors, and hiding important scene details. These degradations pose challenges for various computer vision applications including surveillance systems, intelligent transportation, remote sensing, and outdoor monitoring. The clear image visibility is crucial for accurate analysis and decision-making. To address this issue, our proposed work presents a visibility restoration and image enhancement method for images degraded by atmospheric scattering conditions. Our novel approach is based on the combination of Dark Channel Prior(DCP) technique, with contrast limited adaptive histogram equalization(CLAHE) widely helpful for single-image de-hazing. In this method, the degraded input image is first processed to compute the dark channel, which represents the minimum intensity value within a local image patch and helps estimate the density of haze in the scene. Using the dark channel information, atmospheric light is estimated from the brightest pixels of the dark channel map. Then transmission map is calculated to determine the amount of scene radiance that reaches the camera through the degraded medium. This transmission map is further refined using filtering techniques to obtain smoother and more accurate results. Based on the refined transmission map and estimated atmospheric light, the scene radiance is recovered using the atmospheric scattering model to remove haze effects. Finally, CLAHE is applied to improve image clarity and color distribution. Our proposed results demonstrate effectively to enhance the image visibility which are evaluated by peak signal to noise ratio(PSNR) and structural similarity index measure(SSIM). Among these results, our proposed method shows better clarity as well as in metrics and improved scene details in images affected by fog, smoke, and sand dust conditions.**

***Index Terms*—Image Enhancement, Visibility Restoration, Haze removal, Dark Channel Prior, Fog, Smoke, Sand Dust, Transmission Map, Atmospheric Scattering Model.**

I. Introduction

In recent years, computer vision systems have become a highly advanced field, offering the possibility to use it in autonomous driving, surveillance, remote sensing, robotics and environmental monitoring. These systems are very reliant on the quality of the visual data that is taken using imaging devices. But in a real-life scenario, the quality of the images is usually worsened because of weather conditions, like fog, haze, smoke, as well as sand-dust storms. These conditions decrease the visibility, influence the color and blur fine details, which impacts the work of sight-based systems directly.

One of the most essential issues is visibility degradation, as the perception of the scene directly depends on it. Scattering and absorption of light by atmospheric particles result in loss of contrast, distortion of colors, and blurry images. This complicates the differentiation of objects and decreases

the efficiency of computer vision algorithms. This issue is particularly major in such areas of use as traffic monitoring and autonomous driving, where the proper object detection is critical. Lack of visibility may result in misconceptions of scenes, which may cause bigger problems that lead to accidents. In the same way, low-quality images make it hard to identify any crucial occurrences in the surveillance systems. Thus, it is crucial to improve poor images regarding credibility and security.

The degradation of images in outdoor conditions happens because of the interaction of the atmospheric particles with light. Scattering is a process that focuses the light into a different direction, whereas absorption decreases the intensity of the light. The effect of these conditions may affect low contrast and poor clarity [1]. Moreover, it is also wavelength-dependent, so that attenuation is not evenly distributed over color channels, nor does it create a natural color appearance.

The problem of sand-dust is more complicated than the problems of fog or haze. Dust particles are bigger and irregularly formed thus reflected differently. This causes serious distortion of color and the loss of short wavelengths leads to the production of yellowish or reddish color [2], [3]. It is also a source of noise and blunts sharp edges making it hard to get meaningful information. In addition, due to the asymmetrical nature of particles, spatial diffusion of the degradation in the image causes harder restoration [4].

Considering all these issues of atmospheric conditions, we know that they will mainly affect the images badly with less clarity of features and cannot be used for remote sensing and surveillance applications. In this case, we are mainly focused on avoiding the atmospheric degradations and making better images by using image processing techniques. In this article, we are explaining the literature that belongs to the image processing methods on degraded images, followed by the explanation of the flow of the proposed method, then results and discussion and finally the conclusion.

II. Literature Survey

Machine Learning (ML) and Deep Learning (DL) play a major role in modern image processing. ML techniques use algorithms like support vector machines and decision trees to analyze image features such as edges, textures, and colors for tasks like classification and detection. Deep Learning, a subset of ML, uses neural networks—especially convolutional neural networks (CNNs)—to automatically learn features from images without manual extraction. DL models provide higher accuracy in complex tasks such as object recognition, image segmentation, and enhancement. In image processing, these methods are applied in medical imaging, autonomous vehicles, surveillance, and facial recognition. Compared to traditional methods like Dark Channel Prior, DL approaches can handle diverse conditions and large datasets more effectively, producing robust and adaptive results.

Image visibility restoration has attracted significant attention due to degradation caused by haze, fog, blur, and underwater conditions. Nowadays, researchers explore fusion-based and model-driven approaches to improve performance. Singh et al. developed a fusion-based visibility restoration model optimised using Differential Evolution, which effectively enhances remote sensing images [5]. Liang et al. introduced a generalized image formation model that improves clarity, color fidelity, and detail preservation under poor visibility conditions [6]. Additionally, domain-specific research gained importance, particularly in underwater imaging. Muthuraman et al. proposed a hybrid restoration framework combining neural networks and weighted filtering techniques, while Li et al. introduced a hybrid model integrating physical and data-driven approaches to enhance underwater image quality [7], [8].With the rapid advancement of deep learning, modern techniques have significantly improved image restoration capabilities. CNN-based approaches, such as those proposed by Yasarla and Patel and Cheng et al., effectively address atmospheric turbulence and blur, enhancing fine details and visual quality [9], [10]. Moreover, transformer-based models like MwFormer and advanced generative models such as MACGAN provide robust solutions for multi-weather image restoration, handling complex degradations including haze, fog, and low-light conditions [11], [12]. Recent developments, such as PhyDAE and lightweight dual-view networks, focus on adaptive, efficient, and real-time restoration frameworks suitable for practical applications [13], [14].More about the image processing methods, a major breakthrough in this field was introduced by He et al., who proposed the Dark Channel Prior (DCP) method for single image dehazing. This approach estimates transmission and atmospheric light to effectively recover clear images from hazy inputs [15]. Building upon this, Shuai et al. enhanced the DCP framework by incorporating Wiener filtering, which helps reduce noise amplification and improves the overall restoration quality [16]. Furthermore, Pan et al. extended the concept of dark channel prior to blind image deblurring, demonstrating that it can also restore sharp images without prior knowledge of blur [17].As research evolved, several studies focused on refining and optimising DCP-based methods. Wang et al. improved the traditional DCP technique by addressing its limitations, resulting in faster processing and more natural-looking outputs [18]. In addition, Lee et al. presented a comprehensive review of DCP-based dehazing algorithms, discussing their working principles, advantages, and shortcomings [19]. To further enhance adaptability, Thirumala et al. introduced a haze-level prior approach that dynamically adjusts restoration based on haze intensity [20]. Similarly, Kumari et al. proposed a fast and efficient visibility restoration technique, making it suitable for real-time applications [21]. Furthermore, recent studies emphasize multi-condition and application-specific restoration techniques. Infrared-assisted frameworks for joint restoration and fusion, airport visibility reconstruction models, and sand-dust enhancement methods demonstrate the need for robust and versatile solutions in diverse environments [14], [22], [23]. Comprehensive surveys by Sahu et al. and Alsakar et al. highlight current trends and conclude that, while traditional

methods like DCP remain foundational, deep learning and hybrid approaches offer superior performance, adaptability, and scalability [24], [25].

Overall, the progression from traditional prior-based techniques to advanced deep learning models reflects continuous advancements in image enhancement of atmospherically degraded images. This literature mainly aims to achieve high-quality, efficient, and real-time image restoration under various challenging environmental conditions, making it highly relevant for real-world applications. In this paper, we introduce a novel combination approach of DCP with CLAHE for improved contrast enhancement, along with refined transmission estimation for better edge preservation under complex atmospheric conditions, and it should be suitable for real-time applications.

III. Proposed Method

The proposed work focuses on enhancing the visibility of degraded images using the Dark Channel Prior (DCP) method with the CLAHE framework. Firstly, the DCP method follows a structured pipeline involving estimation of the dark channel, computation of the transmission map, refinement of the transmission, and recovery of scene radiance. These steps collectively address the effects of atmospheric scattering, which causes reduced contrast, color distortion, and loss of detail in images captured under haze, fog, or sand-dust conditions. Then we are applying the CLAHE for computationally efficient, which makes the clarity of details for further applications.

A. Dark Channel Prior (DCP) Method

The Dark Channel Prior (DCP) is a statistical observation that forms the foundation of the proposed method. It states that in most non-sky regions of natural images, at least one color channel such as red, green, or blue, contains pixels with very low intensity values within a local patch. This method is used to estimate the difference between the haze or dehaze conditions in the particular images [19].

Consider an input hazy image $I(x)$, the dark channel $J(x)$ is computed as:

$$I(x) = J(x) \cdot t(x) + A \cdot (1 - t(x)) \tag{1}$$

where:

- $I(x)$ is the observed hazy image,
- $J(x)$ is the scene radiance (desired output),
- $t(x)$ is the transmission map,
- A is the atmospheric light.

Rearranging the equation, the scene radiance can be recovered as:

$$J(x) = \frac{I(x) - A}{\max\left(t_{\text{refined}}(x),\, t_0\right)} + A \tag{2}$$

where:

- t_0 is a lower bound for transmission (typically $t_0 = 0.1$) to prevent division by very small values.

The min operation is applied first across color channels and then across the local patch. The resulting dark channel image highlights regions with dense haze. Bright pixels in the dark channel correspond to areas with heavy atmospheric scattering, while darker regions indicate clearer parts of the image [17].

In addition to the dark channel computation, atmospheric light A is estimated. It represents the global light intensity contributed by the atmosphere. It is typically estimated by selecting the brightest pixels in the dark channel and retrieving their corresponding intensities from the original image:

$$A = \max_{x \in \text{top percentage}} I(x) \tag{3}$$

Accurate estimation of atmospheric light is critical because it directly affects the quality of the restored image. Overestimation may lead to washed-out images, while underestimation can result in insufficient haze removal [17], [19].

B. Initial Transmission Map Estimation

The transmission map represents the portion of light that reaches the camera without being scattered. It is a crucial parameter in the atmospheric scattering model, as it determines the level of visibility at each pixel [26].

Using the Dark Channel Prior, the initial transmission map $t(x)$ is estimated as:

$$t(x) = 1 - \omega \cdot \min_{y \in \Omega(x)} \left(\min_{c \in \{r,g,b\}} \frac{I^c(y)}{A^c} \right) \tag{4}$$

where:

ω is a constant parameter (typically set to 0.95) that controls the amount of haze retained to avoid over-enhancement,

A^c is the atmospheric light in channel c.

This equation normalizes the image with respect to atmospheric light and computes the dark channel of the normalized image. The transmission map obtained represents how much scene radiance is preserved at each pixel.

Lower values of $t(x)$ indicate regions with dense haze, while higher values correspond to clearer regions. However, the initial transmission map may contain block artifacts due to the use of local patches and minimum filtering. These artifacts can degrade the quality of the restored image if not properly addressed [19].

1) Refined Transmission Map Using Guided Filter: To improve the accuracy of the initial transmission map, a refinement process is applied using a guided filter. The guided filter is an edge-preserving smoothing technique that removes noise and block artifacts while maintaining sharp edges.

The refined transmission map $t_{\text{refined}}(x)$ is obtained as:

$$t_{\text{refined}}(x) = \text{GuidedFilter}\big(I(x),\, t(x),\, r,\, \epsilon\big) \tag{5}$$

where:

$I(x)$ is the guidance image (original input image),

$t(x)$ is the initial transmission map,

r is the radius of the filtering window,

ϵ is a regularization parameter controlling smoothness.

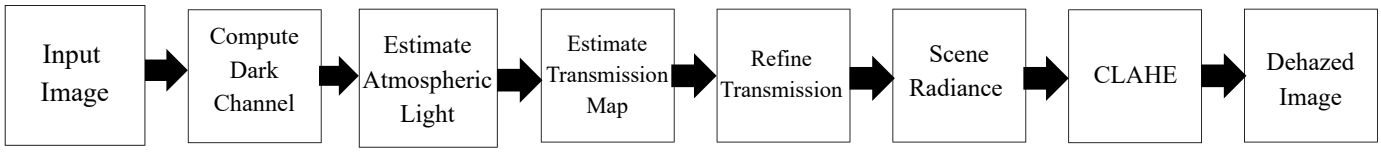

Fig. 1. **Flow chart of the proposed method**

The guided filter works based on a local linear model, where the output is assumed to be a linear transform of the guidance image within a local window. This ensures that edges in the original image are preserved in the refined transmission map [27].

Refinement is necessary because the initial transmission map often suffers from halo artifacts near object boundaries. By applying the guided filter, these artifacts are reduced, and the transmission map becomes more aligned with the true structure of the scene. This leads to improved restoration quality in the final image [28].

C. Scene Radiance Recovery

After obtaining the refined transmission map and estimating atmospheric light, the final step is to recover the scene radiance, which represents the true appearance of the scene without atmospheric degradation.

The atmospheric scattering model is given by:

$$I(x) = J(x) \cdot t(x) + A \cdot \big(1 - t(x)\big) \tag{6}$$

where:
$I(x)$ is the observed hazy image,
$J(x)$ is the scene radiance (desired output),
$t(x)$ is the transmission map,
A is the atmospheric light.

Rearranging the equation, the scene radiance can be recovered as:

$$J(x) = \frac{I(x) - A}{\max\big(t_{\text{refined}}(x),\, t_0\big)} + A \tag{7}$$

where:
t_0 is a lower bound for transmission (typically 0.1) to prevent division by very small values.

The use of t_0 ensures numerical stability and prevents excessive amplification of noise in regions with very low transmission. This step effectively removes the haze component and restores the original scene contrast and color.

After recovering the scene radiance, a clipping operation is applied to ensure that pixel values remain within a valid range [29]. This is followed by contrast enhancement to improve the visual appearance of the image. The final output image exhibits improved clarity, enhanced contrast, and restored color balance [30].

D. Contrast Limited Adaptive Histogram Equalization

CLAHE (Contrast Limited Adaptive Histogram Equalization) is a technique used to improve the contrast of an image in a smart way. Instead of enhancing the whole image at once, it divides the image into small parts (tiles) and improves each part separately. It also uses a limit (clipping) to avoid over-brightening and noise. Because of this, CLAHE enhances details without making the image look unnatural.

The importance of CLAHE is that it makes images clearer and more visually appealing by improving brightness, contrast, and fine details. In your project, after removing haze using the DCP method, CLAHE helps to further enhance the image quality by making hidden details more visible and improving overall clarity [31].

E. Summary of Proposed Work

The proposed method integrates the Dark Channel Prior with transmission estimation, guided filtering, and scene radiance recovery, along with CLAHE, to enhance degraded images effectively. Each stage of the process addresses specific challenges associated with atmospheric degradation, such as noise degradation and edge enhancement. The DCP method identifies haze regions, the transmission map quantifies visibility, the guided filter refines structural details, and the radiance recovery step reconstructs the clear image. For further enhancement of details and improved edge clarity, CLAHE is used for contrast improvement.

Overall, the proposed approach provides a robust and computationally efficient solution for image visibility enhancement. This method successfully improves contrast, reduces haze effects, and preserves important image features, making it suitable for real-world applications such as autonomous driving, surveillance, and outdoor vision systems.

IV. Results and Discussions

The proposed work provides a comprehensive assessment of the proposed strategy for improving the visibility of sand-dust, fog, and smoke-degraded images, including qualitative and quantitative analysis evaluations. Here we are using some of the atmospheric degraded images, which are from Kaggle datasets []. We performed our simulations using MATLAB software. The degraded and ground truth images, which are depicted as Fig. 2and Fig. 3 This study comprehensively compares our proposed method with all previous methods [30], [32]–[34]. To ensure a fair comparison, all conventional image processing and dictionary learning methods were evaluated

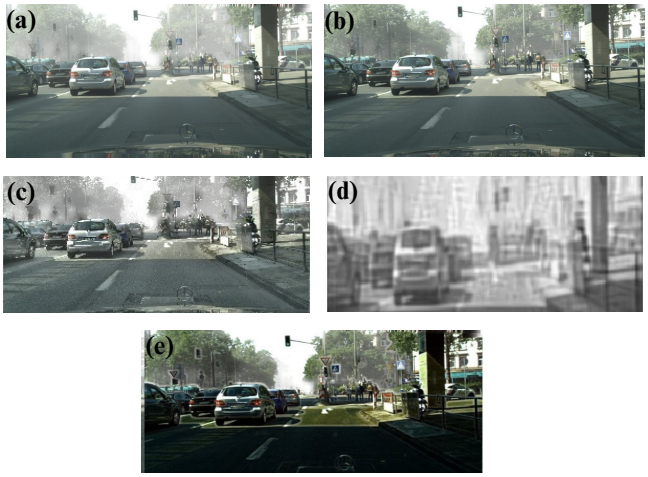

Fig. 2. **Comparative results of Image 1 are (a)Degraded Input image 1, (b)Ground Truth, (c) Non local filter with CLAHE, (d) k-SVD with CLAHE, (e) DCP method with CLAHE**

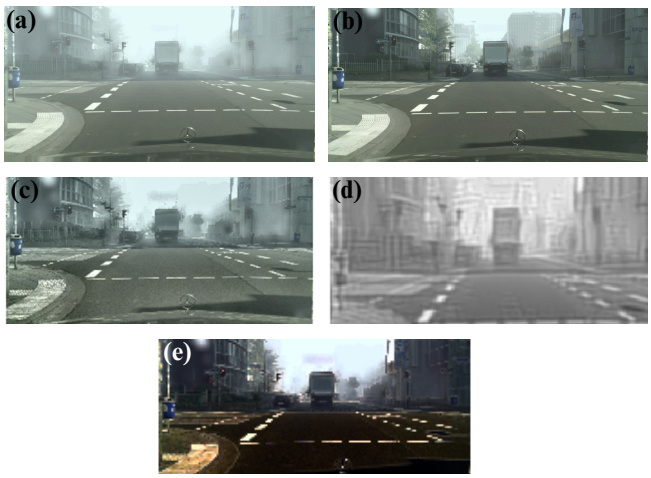

Fig. 3. **Comparative results of Image 2 are (a))Degraded Input image 2, (b)Ground Truth, (c)Non local filter with CLAHE, (d)k-SVD with CLAHE, (e)DCP method with CLAHE**

both qualitatively and quantitatively. For more clarity, we evaluate our results using quantitative metrics such as PSNR and SSIM values, which are very important for proper image enhancement. In our work, we compare our proposed method with a conventional image processing method as a combination of denoising and enhancement using a nonlinear filter with CLAHE, which is depicted in Fig. 2(c) and Fig. 3(c). From these figures, it is illustrated that the NLM with the CLAHE method improves color distortion but still causes blurring of edge details. To overcome this blurring effect, we explore the dictionary learning technique, which is the K-SVD (K-Singular Value Decomposition) method for efficient denoising. In this paper, for comparison, we incorporate K-SVD with CLAHE, and the results are demonstrated in Fig. 2(d) and Fig. 3(d). Similarly, we observe the same noise and blurring effect in the edge details in this case as well. These limitations in addressing complex sand-dust formations and foggy images result in inadequate visual restoration and quantitative performance.

Later, haze image processing is compared with the CNN-based approach [9]. However, it has difficulty handling thick and uneven haze, leading to inconsistent results. The image quality and edge details are not satisfactory.

Finally, we apply our proposed framework, which is a combination of the Dark Channel Prior with CLAHE. This approach is practical in approximating the initial transmission map, and the accuracy and edge retention are enhanced by refining it further [19], as depicted in Fig. 2(e) and Fig. 3(e). The proposed method is strong and consistent under varying haze conditions. The quantitative analysis of these methods is tabulated and shown in Table. IV and Table. IV. From the

TABLE I
QUANTITATIVE ANALYSIS OF HYBRID COMBINATION OF DENOISING AND ENHANCED METHODS ON IMAGE 1

Methods	***PSNR***	***SSIM***
Non-Linear Filter with CLAHE	11.04	0.4512
k-SVD with CLAHE	13.56	0.5842
DCP Method with CLAHE	19.82	0.7914

TABLE II
QUANTITATIVE ANALYSIS OF HYBRID COMBINATION OF DENOISING AND ENHANCED METHODS ON IMAGE 2

Methods	***PSNR***	***SSIM***
Non-Linear Filter with CLAHE	11.23	0.4215
k-SVD with CLAHE	15.42	0.6841
DCP Method with CLAHE	18.95	0.7432

tables, we can clearly see that our proposed method shows better PSNR and SSIM metrics compared to the conventional and dictionary learning methods. These better metrics indicate improved edge enhancement, which is necessary for appropriate visual analysis in real-time systems. The results show that the proposed method is superior in retaining anisotropy and improving features across all channels (RGB), making it especially effective for improving multi-spectral visibility.

CONCLUSION

In conclusion, the proposed visibility restoration framework effectively addresses image degradation caused by atmospheric conditions such as fog, smoke, and sand dust. By utilizing the Dark Channel Prior with the CLAHE method which accurately estimates the dark channel, atmospheric light, and transmission map to recover scene radiance. The refinement of the transmission map ensures smoothness while preserving important structural details, leading to reduced artifacts and improved visual quality. Furthermore, the application of CLAHE enhances image clarity and color distribution. Our results confirm that our proposed approach significantly improves visibility, contrast, and detail preservation by evaluating the quantitative metrics such as PSNR and SSIM. Owing to its efficiency and

simplicity, our proposed method is better in metrics and well-suited for practical applications in surveillance, transportation, remote sensing, and day-to-day life scenarios.

REFERENCES

[1] Y. Wang and D. Durmus, "The effect of color contrast on the visual clarity of images of complex indoor environments," *Buildings*, vol. 15, no. 7, p. 1157, 2025.

[2] Y. Cheng, Z. Jia, H. Lai, J. Yang, and N. K. Kasabov, "A fast sand-dust image enhancement algorithm by blue channel compensation and guided image filtering," *IEEe Access*, vol. 8, pp. 196 690–196 699, 2020.

[3] M. K. K. Masood, E. N. Baro, and P. O. Roth, "A novel intensity-corrected blue channel compensation and edge-preserving contrast enhancement using laplace filter and sigmoid function for sand-dust image enhancement," *IEEE Access*, vol. 13, pp. 43 127–43 144, 2025.

[4] J. E. Santos, A. Marcato, R. Colman, N. Lubbers, and Y. T. Lin, "Discrete spatial diffusion: Intensity-preserving diffusion modeling," *arXiv preprint arXiv:2505.01917*, 2025.

[5] D. Singh, M. Kaur, M. Y. Jabarulla, V. Kumar, and H.-N. Lee, "Evolving fusion-based visibility restoration model for hazy remote sensing images using dynamic differential evolution," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–14, 2022.

[6] Z. Liang, W. Zhang, R. Ruan, P. Zhuang, and C. Li, "Gifm: An image restoration method with generalized image formation model for poor visible conditions," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–16, 2022.

[7] D. L. Muthuraman and S. M. Santhanam, "Visibility improvement of underwater turbid image using hybrid restoration network with weighted filter," *Multidimensional Systems and Signal Processing*, vol. 33, no. 2, pp. 459–484, 2022.

[8] X. Li, G. Hou, L. Tan, and W. Liu, "A hybrid framework for underwater image enhancement," *IEEe Access*, vol. 8, pp. 197 448–197 462, 2020.

[9] R. Yasarla and V. M. Patel, "Cnn-based restoration of a single face image degraded by atmospheric turbulence," *IEEE Transactions on Biometrics, Behavior, and Identity Science*, vol. 4, no. 2, pp. 222–233, 2022.

[10] J. Cheng, W. Zhu, J. Li, G. Xu, X. Chen, and C. Yao, "Restoration of atmospheric turbulence-degraded short-exposure image based on convolution neural network," in *Photonics*, vol. 10, no. 6. MDPI, 2023, p. 666.

[11] R. Zhu, Z. Tu, J. Liu, A. C. Bovik, and Y. Fan, "Mwformer: Multi-weather image restoration using degradation-aware transformers," *IEEE Transactions on Image Processing*, vol. 33, pp. 6790–6805, 2024.

[12] M. Siddiqua, S. B. Belhaouari, N. Akhter, A. Zameer, and J. Khurshid, "Macgan: An all-in-one image restoration under adverse conditions using multidomain attention-based conditional gan," *IEEE Access*, vol. 11, pp. 70 482–70 502, 2023.

[13] Z. Dong, Z. Zhang, Y. Sun, H. Jiang, T. Liu, and Y. Gu, "Phydae: Physics-guided degradation-adaptive experts for all-in-one remote sensing image restoration," *IEEE Transactions on Geoscience and Remote Sensing*, 2026.

[14] G. Gao, Y. Xiao, X. Wen, C. Sun, and Y. Wang, "A lightweight dual-view network for sand-dust degraded image enhancement," *Knowledge-Based Systems*, p. 115308, 2026.

[15] K. He, J. Sun, and X. Tang, "Single image haze removal using dark channel prior," *IEEE transactions on pattern analysis and machine intelligence*, vol. 33, no. 12, pp. 2341–2353, 2010.

[16] Y. Shuai, R. Liu, and W. He, "Image haze removal of wiener filtering based on dark channel prior," in *2012 Eighth international conference on computational intelligence and security*. IEEE, 2012, pp. 318–322.

[17] J. Pan, D. Sun, H. Pfister, and M.-H. Yang, "Blind image deblurring using dark channel prior," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 1628–1636.

[18] X. Wang, Z. Chu, and Z. Li, "Optimization research on single image dehazing algorithm based on improved dark channel prior," *Artificial Intelligence and Machine Learning Review*, vol. 4, no. 4, pp. 57–74, 2023.

[19] S. Lee, S. Yun, J.-H. Nam, C. S. Won, and S.-W. Jung, "A review on dark channel prior based image dehazing algorithms," *EURASIP Journal on Image and Video Processing*, vol. 2016, no. 1, p. 4, 2016.

[20] V. L. Thirumala, V. S. Karanam, P. R. Lankireddy, A. K. Kakumani, and R. K. Yacharam, "Haze-level prior approach to enhance object visibility under atmosphericdegradation," *Turkish Journal of Electrical Engineering and Computer Sciences*, vol. 29, no. 2, pp. 994–1014, 2021.

[21] A. Kumari, S. K. Sahoo, and M. Chinnaiah, "Fast and efficient visibility restoration technique for single image dehazing and defogging," *IEEE Access*, vol. 9, pp. 48 131–48 146, 2021.

[22] Y. Zhang, Y. Liu *et al.*, "Infrared-assisted single-stage framework for joint restoration and fusion of visible and infrared images under hazy conditions," *Pattern Recognition*, p. 113074, 2026.

[23] F.-w. Liu and R.-j. Wu, "An intelligent hybrid machine learning framework for low-visibility reconstruction for airport transportation," *Signal, Image and Video Processing*, vol. 20, no. 3, p. 170, 2026.

[24] G. Sahu, A. Seal, D. Bhattacharjee, M. Nasipuri, P. Brida, and O. Krejcar, "Trends and prospects of techniques for haze removal from degraded images: A survey," *IEEE Transactions on Emerging Topics in Computational Intelligence*, vol. 6, no. 4, pp. 762–782, 2022.

[25] Y. M. Alsakar, N. A. Sakr, S. El-Sappagh, T. Abuhmed, and M. Elmogy, "Underwater image restoration and enhancement: a comprehensive review of recent trends, challenges, and applications," *The Visual Computer*, vol. 41, no. 6, pp. 3735–3783, 2025.

[26] S. E. Kim, T. H. Park, and I. K. Eom, "Fast single image dehazing using saturation based transmission map estimation," *IEEE Transactions on Image Processing*, vol. 29, pp. 1985–1998, 2019.

[27] S. M. Ehsan, M. Imran, A. Ullah, and E. Elbasi, "A single image dehazing technique using the dual transmission maps strategy and gradient-domain guided image filtering," *IEEe Access*, vol. 9, pp. 89 055–89 063, 2021.

[28] L. Rahmawati, S. Rustad, A. Marjuni, M. Arief, C. S. Soeleman, and G. F. Shidik, "Transmission map refinement using laplacian transform on single image dehazing based on dark channel prior approach," *Cybern. Inf. Technol*, vol. 24, no. 4, 2024.

[29] P. E. Debevec and J. Malik, "Recovering high dynamic range radiance maps from photographs," in *Seminal Graphics Papers: Pushing the Boundaries, Volume 2*, 2023, pp. 643–652.

[30] S. Venkatesh, C. J. De Britto, P. Subhashini, and K. Somasundaram, "Image enhancement and implementation of clahe algorithm and bilinear interpolation," *Cybern. Syst*, vol. 2022, pp. 1–13, 2022.

[31] R. C. Gonzalez, *Digital image processing*. Pearson education india, 2009.

[32] X. Liu, Z. Wu, and X. Wang, "Validity of non-local mean filter and novel denoising method," *Virtual reality & intelligent hardware*, vol. 5, no. 4, pp. 338–350, 2023.

[33] R. Chen, D. Pu, Y. Tong, and M. Wu, "Image-denoising algorithm based on improved k-singular value decomposition and atom optimization," *CAAI Transactions on Intelligence Technology*, vol. 7, no. 1, pp. 117–127, 2022.

[34] Y. Al Najjar, "Comparative analysis of image quality assessment metrics: Mse, psnr, ssim and fsim," *International Journal of Science and Research (IJSR)*, vol. 13, no. 3, pp. 110–114, 2024.